AUtoMLy | Medicaid Policy Copilot

INAI 6986 — Milestone 1: Data and Model Selection

J.O. • Dr. David M. Simmonds • Evidence through September 30, 2026, 12:34 PM CDT

Decision and Evidence Boundary

The original September 24 evidence remains historical; later work is a post-feedback supplement, not evidence available at the original deadline. The current experiment is the completed nested N=200/509/750/1,600 series for both Gemma and Muse on shared prompt IDs and fixed validation. September 16–18 Muse runs used earlier methodology and are excluded from current result tables and plots; they remain archived only as historical provenance. Tokenizer-specific fit metrics cannot rank factual answers across families. No model has been promoted. The protected final control/test remain unspent; human leakage review and a reasoning/abstention control remain open.

1. Data Characteristics

The task is a source-cited answer to a public Alabama Medicaid policy or provider-billing question, with a source locator or an abstention when support is absent. It is not an automated eligibility, payment, clinical or legal decision. Each candidate retains source URL, document identity, PDF page, snapshot date and SHA-256 file fingerprint. The target is a response grounded in the frozen source, not a class label.

Prepared Evidence	Measured Shape / Consequence
Source snapshot	138 public PDF files / 2,433 pages (September 14); 136 yielded candidate records.
Ordinary examples	1,999 answerable supervised fine-tuning (SFT) records; a separate one-record intentionally unanswerable case was quarantined.
Source mix	750 Administrative Code and 1,249 Provider Manual records. Median answer length 18 whitespace-delimited words; 95th percentile 52.
Historical clean split	1,684 train / 124 validation / 191 historical test records; 113 / 10 / 13 distinct source documents, respectively; zero document overlap.
Rights and privacy	Approved public information only; no patient claims, protected health information or agency-confidential records.

Preparation and Revised Split

The document-grouped split assigns every page of a source PDF together, preventing page-neighbor leakage. One blank-page safety example was removed from ordinary SFT. Earlier September 16–18 Muse experiments used pre-amendment generator, split and training procedures. They are retained in the historical archive for provenance only; none of their run metrics is pooled into the current comparison, tables or plots.

The new experiment nests training subsets of 200, 509, 750 and 1,600 from the same 1,684 clean training rows, covers all 113 training documents even at 200, and holds the same 124 validation rows. Earlier proposals for separate 100/250/200/300-record configuration tests were deferred to protect one final test event. The already exposed historical 191 are retrospective diagnostics, not an untouched final test. The thirty hand-authored control questions remain locked; all their source documents are in the historical test group, none in train or validation. Their question wording and keys are independently authored, but their documents are not independent of the historical test documents.

2. Candidates Considered

The declared family is deep learning: local language-model adaptation. Instructor permission covers fine-tuning but not a weakened evaluation standard. The current matched training comparison includes four nested sizes each for Gemma and Muse; the earlier September 16–18 Muse outputs are historical only and are excluded from current results. The revised within-family alternatives differ in base architecture, memory demands, tokenizer and deployment feasibility; an eligible training smoke is not a model-quality result.

Approach	Why Plausible / Trade-Off	Verified Status
Muse Glimmer 30B, rank-16 low-rank adapter	Four-bit local baseline; fresh target-specific render and recipe (rank 16, LR 2×10^{-4}).	Four current nested-size runs completed Sep. 29–30 on the same train prompt IDs and 124 validation IDs as Gemma; earlier runs are archived only.
google/gemma-4-26B-A4B-it, rank-4 adapter	Different mixture-of-experts architecture and tokenizer; token loss/accuracy cannot rank factual answers across model families.	N=200, N=509, N=750 and N=1,600 project-data runs complete. N=1,600 stopped at step600/epoch3; best validation checkpoint400. Same fixed validation; no test/control rows used.
nvidia/NVIDIA-Nemotron-3.5-Lightning-30B-A3B-BF16	Official Megatron Bridge recipes document BF16 LoRA on H100/GB200; local RTX PRO/A6000 backward and checkpoint support is not proven.	Architecture-family documentation reviewed; exact local memory/backward/checkpoint gate remains open; no project-data training.
ornith-ai/Ornith-1.5-35B-A3B	Qwen3.5 MoE family; family-level LoRA memory guidance does not prove this exact checkpoint fits or trains locally.	Exact backward/checkpoint support and local memory fit remain unverified; no project-data training.

3. Baseline

The current-practice baseline is the student manually finding a source and writing an answer. On thirty simple, hand-authored code/page lookups, all thirty gold answers were source-validated; retrospectively paired session timestamps estimate 42:06 total (about 1:24 each). These are easy, non-blind lookups, not a stopwatch trial and not proof of time savings on policy reasoning. Historical Muse pilot outputs are retained in a separate archive and are excluded from this current matched experiment. Neither baseline has yet been measured on the proposed harder control.

Why the Original Control Cannot Decide the Winner

All 30 items ask an answerable page or code lookup. Ten come from Jul26_34.pdf page 2, six from Jul26_28.pdf page 2 and five from Apr26_37.pdf page 2: 21/30 on three pages. This is an honest lookup diagnostic, not a representative policy-support or abstention benchmark. Eight to ten plausible unsupported cases and student-authored exception/date/conflict/multi-rule questions are required before an expanded control is frozen; none has been supplied or scored in this document.

4. Evidence and Evaluation

The committed primary outcome is human-adjudicated factual accuracy on answerable questions; unsupported-claim rate, correct locator and abstention are safety-critical secondary outcomes. No human-scored comparison on the independent hand-authored control or expanded reasoning/abstention set has occurred. The table below reports only the current matched Gemma/Muse nested-size runs; all values are teacher-forced next-token fit diagnostics, not answer-level accuracy.

Historical Muse experiments from September 16–18 used prior methodology. Their metrics are not displayed in the current results table or plots and remain in the archive only for historical provenance.

Current Run	Final Step / Epoch	Final Train Loss / Next-Token Acc.	Best Validation Step / Loss / Acc.	Final Validation Loss / Acc.
Gemma; N=200	75 / 3.00	0.6893 / 84.40%	75: 0.7044 / 83.45%	0.7044 / 83.45%
Gemma; N=509	320 / 5.00	0.3373 / 90.81%	128: 0.6060 / 84.93%	0.6636 / 85.03%
Gemma; N=750	376 / 4.00	0.2192 / 93.62%	188: 0.5829 / 85.58%	0.6946 / 85.13%
Gemma; N=1,600	600 / 3.00	0.2435 / 93.05%	400: 0.5750 / 85.88%	0.6180 / 85.65%
Muse; N=200	75 / 3.00	0.2376 / 93.15%	25: 0.5853 / 85.10%	0.6589 / 84.54%
Muse; N=509	192 / 3.00	0.0856 / 97.64%	64: 0.5727 / 85.44%	0.7456 / 84.70%
Muse; N=750	282 / 3.00	0.0846 / 97.70%	94: 0.5725 / 85.44%	0.7447 / 84.80%
Muse; N=1,600	600 / 3.00	0.0877 / 97.56%	200: 0.5787 / 85.54%	0.7249 / 84.72%

The nested sizes are 200, 509, 750 and 1,600 training rows with the same 124 validation IDs per size; train membership is matched by prompt ID across the model families. Each run completed with zero protected test/control rows. No answer-level validation accuracy is reported. Gemma N=1,600 finished at step 600/epoch 3 with best validation checkpoint 400; Muse N=1,600 finished at step 600/epoch 3 with best checkpoint 200. The full epoch-boundary record, including scored-token denominators and the N=509 resume boundary, appears in Appendix E. Different model tokenizers/recipes mean loss and next-token accuracy should be interpreted within each family, not used to rank cross-family factual accuracy. The one-time final test is still unused.

5. Selection and Justification

No candidate is selected or promoted. The current evidence consists of completed matched nested Muse and Gemma training runs at N=200/509/750/1,600, with model-specific tokenization and recipes; their token loss and next-token accuracy are within-family training/validation diagnostics, not cross-family factual-answer rankings. The September 16–18 Muse runs remain in the historical archive only and do not appear in the current results table or plots. A human-adjudicated answer-level comparison has not been authorized on validation and has not occurred; do not substitute the protected test or control to choose a winner. At the cutoff, Nemotron and Ornith still have no project-data training results and require exact local training gates. Only after an independently defensible candidate selection, dataset lock and human leakage disposition can the one-time final evaluation compare it against untouched base.

6. Interpretability and Leakage Check

The practical interpretability check for a language model is whether answers depend on the right exception, effective date, negation and source locator—not merely whether it emits a URL. No paired ablation has been run. Structural checks found zero document overlap, two duplicate-question groups with no cross-split match, one train–validation near-duplicate question candidate at Jaccard ≥0.85, and sixteen cross-split duplicate-answer groups affecting fifty-seven records. The group/record counts are reproduced from the canonical file with the project's punctuation-insensitive normalization; Appendix C lists each group. Identical short codes may be coincidental; repeated substantive policy text or template language may transmit information across split boundaries. Neither possibility is a human disposition.

Open Leakage and Quality Gate

The student must review all 16 answer groups against both questions and source pages, decide which are harmless shared values, boilerplate, or substantive repeated content, and remove/re-group any contaminated held-out item before a clean comparative score. The near-duplicate question pair also needs a source-level ruling. Existing LLM-scored workbook examples are not a human sign-off. A flagged H2011 "Daily" table-column error requires human verification and correction/quarantine before any affected row is reused.

Every existing control item maps by full source URL and SHA-256 to a historical test document; Appendix B makes the per-item result visible. That refutes the specific concern that these thirty questions draw on training documents in the clean lineage, but does not fix their simple wording, source-page concentration, possible generated-test document overlap or absence of unanswerable cases. A revised student-authored control should be frozen separately and excluded from training, development tuning and candidate choice. Plan a sufficiently varied, source-verified answerable stratum and eight to ten plausible-but-unsupported items; score abstention and locator correctness with explicit denominators and a human adjudication key. Do not call the old 1/1 safety probe a 90% estimate.

Attribution Is a Later Experiment

Open WebUI and ChromaDB retrieval have not been built and deployed for this comparison. No retrieval benefit is claimed. Model-only control scoring can follow a legitimate selection and dataset lock; paired retrieval-on versus retrieval-off attribution waits for the later deployed user interface. This dependency does not excuse unresolved leakage or missing model-only policy-support evidence.

7. What Would Change My Mind

A candidate would be rejected or reconsidered if independently scored, source-verified policy questions do not improve answerable factual accuracy by at least 10 percentage points against untouched base, locator correctness is under 95%, unsupported claims reach 2% or more, abstention is under 90% on a meaningful unsupported set, safety routing is under 95%, general/format regression exceeds 3 points, interactive latency fails, or any leakage or rights defect persists. These are project targets, not results. A strong next-token score or a properly formatted URL cannot override human-scored factual and safety failures.

The October 9, 5:00 PM CDT fallback remains a decision gate: if an adapter-loaded or merged model has not produced a complete validated answer, stop adaptation on the critical path and pursue untouched base plus versioned retrieval, reporting adaptation as a negative result. If retrieval later supplies nearly all of the benefit, credit retrieval rather than fine-tuning.

8. Repository and Reproducibility

Public course deliverables: github.com/porespellar/INAI6986_project_deliverables

Archived earlier-phase notebook (not the current model-results source): milestone1_reproducibility.ipynb

That executed notebook documents the original dataset and legacy diagnostics; its historical model outputs are not part of the current comparison. The current Gemma/Muse run matrix in Section 4 and the 28-row epoch record in Appendix E are generated from artifacts/model_comparison/2026-09-25/results.jsonl; run-level provenance is in run_log.csv and the per-run completion artifacts. These current results use no protected test or control rows. The public repository hosts only the identity-protected J.O. deliverables.

Evidence Gates Before Comparative Scoring

Gate	Current Disposition / Required Action
Source membership	Closed for original 30: 30/30 on historical test documents; zero on clean train/validation. Preserve hash-matched item mapping.
Question validity	Open: student authors exception, effective-date, conflict and multi-rule items across more sources; original 30 remain lookup-only.
Abstention	Open: add 8–10 independently authored unsupported questions and source-verify the bounded absence of an answer.
Duplicate/quality flags	Open: human-dispose all 16 groups / 57 records and one near-duplicate question pair; resolve flagged H2011 row.
Candidate choice	Open: train/validation token metrics alone cannot rank cross-tokenizer factual generalization; do not peek at protected control.
Final model test	Not run: lock dataset, recipe, checkpoint, comparator and scorer first; evaluate the frozen candidate and untouched base once.

This supplement addresses the feedback with verified repairs and an explicit unfinished-work boundary. It is not a claim that the expanded control, human leakage disposition, final comparison, or a particular course score has been achieved.

Appendix A — AI Use Disclosure

AI assistance. Hermes Agent (GPT-6 Sol 900k for the initial review and September 30 evidence refresh; GPT-6 Luna 900k for the earlier update) reviewed the professor’s feedback, rubric, saved project evidence and document layout; drafted and validated this supplement and its build code. Earlier local Muse generated original SFT candidates. AI-suggested leakage categories and document prose are not human source adjudication, independently authored control questions, or model outcomes.

Human authorship and accountability. J.O. conceived the project and selected the architecture, source scope, model candidates, metrics, split and guarded workflow; configured local hardware and services, directed agents, diagnosed training and checkpoint failures, and made proceed/stop/quarantine decisions. J.O. wrote and source-verified the original thirty control questions and answer keys. J.O. must author any new control questions, review flagged source evidence and approve final conclusions before submission. The student retains final responsibility for all content and project actions.

Verification. The builder asserts the clean dataset hash, all 30 historical test-document mappings, the 16-group/57-record duplicate count, all four Gemma and all four Muse nested-size runs completed, shared train/validation prompt IDs per size, 28 epoch-level records, and zero test/control rows. Legacy September 16–18 Muse records are excluded from the current result matrix. DOCX/PDF checks are delivery checks, not human adjudication.

Appendix B — Independent Control Source-Split Map

The original source-verified, student-authored set contains thirty answerable lookup questions from ten PDF documents and eleven document/page locations. The three most concentrated pages account for 21/30. Mapping uses full source URL plus SHA-256 against the canonical f0d9324c68e7865f87cb1da3351d059b25ed08d2e08b9548073860ca7c9f702c dataset; every listed source is in the historical test split. The original questions are preserved verbatim; their source screenshot records remain in the original submission and local evidence directory. The table is a membership audit, not an expanded control or a model score.

ID	Original Student-Authored Question	Expected Answer	Source PDF / Page	Split
SA-001	What page number in the table of contents lists Rule 560-X-1-.07, Provider Rights and Responsibilities?	Page 3.	9.2_Adm_Code_Chap_1_General_3-18-21.pdf p.1	test
SA-002	What page number in the table of contents lists Rule 560-X-28-.02, Summary of State Law Regarding Advanced Directives?	Page 2.	9.2_Adm_Code_Chap_28_Alabama_Medicaid_Forms_7-26-18.pdf p.1	test
SA-003	What page number in the table of contents lists Rule 560-X-49-.07, Copayment?	Page 2.	9.2_Adm_Code_Chap_49_Certified_NP_Program_9-23-19.pdf p.1	test
SA-004	What page number in the table of contents lists Rule 560-X-50-.03, Recipient Eligibility?	Page 3.	9.2_Adm_Code_Chap_50_Preventive_Health_Ed_Program.pdf p.1	test
SA-005	What Medicaid provider type is assigned to Licensed Social Workers?	Provider type 11, Crossovers Only.	Apr26_23.pdf p.1	test
SA-006	What provider type is assigned to therapy providers?	Provider type 17, Therapy.	Apr26_37.pdf p.2	test
SA-007	What provider type is assigned to Remote Patient Monitoring providers?	Provider type 08, Remote Patient Monitoring.	Jul26_111.pdf p.1	test
SA-008	What provider type is assigned to physicians?	Provider type 31, Physician.	Jul26_28.pdf p.2	test
SA-009	What provider type is assigned to preventive health education providers?	Provider type 55.	Jul26_30.pdf p.1	test
SA-010	What provider type is assigned to psychology providers?	Provider type 54, Psychologist.	Jul26_34.pdf p.2	test
SA-011	What provider type is assigned to Behavioral Health providers?	Provider type 07, Behavioral Health.	Jul26_34.pdf p.2	test
SA-012	What specialty code is listed for Psychology providers?	Psychology specialty code 112.	Jul26_34.pdf p.2	test
SA-013	What specialty code is listed for a Licensed Professional Counselor / LPC?	Specialty code 070	Jul26_34.pdf p.2	test
SA-014	What provider type is assigned to Nurse Practitioners?	Provider type 09.	Jul26_28.pdf p.2	test
SA-015	What provider type is assigned to Physician Assistants?	Provider type 10.	Jul26_28.pdf p.2	test

Control Source-Split Map — Continued

ID	Original Student-Authored Question	Expected Answer	Source PDF / Page	Split
SA-016	What specialty code is listed for Occupational Therapy?	Occupational Therapy code 171.	Apr26_37.pdf p.2	test
SA-017	What specialty code is listed for Physical Therapy?	Physical Therapy code 170.	Apr26_37.pdf p.2	test
SA-018	What specialty code is listed for Speech Therapy?	Speech Therapy code 173.	Apr26_37.pdf p.2	test
SA-019	What specialty code is listed for Applied Behavior Analysis Therapy.	Applied Behavior Analysis (ABA) Therapy. Code 175.	Apr26_37.pdf p.2	test
SA-020	What provider specialties are assigned to Remote Patient Monitoring providers?	Provider specialties 085 or 086.	Jul26_111.pdf p.1	test
SA-021	What specialty code is listed for an Associate Licensed Counselor / ALC?	Specialty code 071.	Jul26_34.pdf p.2	test
SA-022	What specialty code is listed for a Licensed Marriage and Family Therapist / LMFT?	Specialty code 072.	Jul26_34.pdf p.2	test
SA-023	What specialty code is listed for a Licensed Marriage and Family Therapist Associate / LMFTA?	Specialty code 075.	Jul26_34.pdf p.2	test
SA-024	What specialty code is listed for a Licensed Master Social Worker / LMSW?	Specialty code 073.	Jul26_34.pdf p.2	test
SA-025	What specialty code is listed for a Licensed Independent Clinical Social Worker / LICSW.	Specialty code 074.	Jul26_34.pdf p.2	test
SA-026	What specialty code is listed for a Licensed Psychological Technician / LPT?	Specialty code 076.	Jul26_34.pdf p.2	test
SA-027	What specialty code is listed for Preventive Health Education?	Specialty code 183.	Jul26_30.pdf p.1	test
SA-028	What specialty code is listed for EPSDT in the physician specialty table?	EPSDT code 560.	Jul26_28.pdf p.2	test
SA-029	What specialty code is listed for Family Practice in the physician specialty table?	Family Practice code 316.	Jul26_28.pdf p.2	test
SA-030	What specialty code is listed for Cardiac Surgery in the physician specialty table?	Cardiac Surgery code 312.	Jul26_28.pdf p.2	test

Source concentration (document/page → number of questions): Jul26_34.pdf p.2: 10; Jul26_28.pdf p.2: 6; Apr26_37.pdf p.2: 5; Jul26_111.pdf p.1: 2; Jul26_30.pdf p.1: 2; 9.2_Adm_Code_Chap_1_General_3-18-21.pdf p.1: 1; 9.2_Adm_Code_Chap_28_Alabama_Medicaid_Forms_7-26-18.pdf p.1: 1; 9.2_Adm_Code_Chap_49_Certified_NP_Program_9-23-19.pdf p.1: 1; 9.2_Adm_Code_Chap_50_Preventive_Health_Ed_Program.pdf p.1: 1; Apr26_23.pdf p.1: 1.

Appendix C — Cross-Split Answer-Overlap Audit

These are reproducible flags, not human rulings. The canonical file yields sixteen punctuation/case-normalized answer groups spanning splits (57 member records). Review the questions, cited source pages and semantic meaning for every group in the human review workbook; a shared page number, repeated boilerplate and repeated policy answer require different remedies. Do not delete records merely because an answer string repeats. Keep the frozen original file and version a corrected split/file after human approval.

Answer Group (abbreviated)	Members	Splits	Provisional Risk / Action
page 2	8	test/train/validation	Needs source-level human ruling
this page intentionally left blank	7	test/train/validation	Needs source-level human ruling
the current procedural terminology cpt and current dental termin…	6	train/validation	Needs source-level human ruling
page 3	5	test/train/validation	Needs source-level human ruling
3 90	4	test/train	Needs source-level human ruling
icd 9 codes should be used for claims submitted with dates of se…	4	test/train/validation	Needs source-level human ruling
3	3	test/train	Needs source-level human ruling
4	3	train/validation	Needs source-level human ruling
chapter 5	3	train/validation	Needs source-level human ruling
11	2	test/train	Needs source-level human ruling
334 215 7416	2	test/train	Needs source-level human ruling
a hard copy ub 04 claim form must be submitted	2	train/validation	Needs source-level human ruling
the amount billed shall not exceed the entity s actual acquisiti…	2	test/train	Needs source-level human ruling
the current procedural terminology cpt and current dental termin…	2	train/validation	Needs source-level human ruling
the ltc gainwell cover sheet from the web portal must accompany …	2	train/validation	Needs source-level human ruling
up to four procedure code modifiers	2	test/train	Needs source-level human ruling

The one cross-split near-duplicate question pair remains open. The separate quality-review human template and AI example are in artifacts/milestone1/quality_review/2026-09-24/; no sign-off is inferred from their existence. The original generated-set thirty-card human factual pilot is a different activity.

Appendix D — Local Evidence Ledger and Sources

Claim / Requirement	Verifiable Artifact
Rubric and instructor's scored feedback	Assignment 3: Milestone 1 — Data & Model Selection, course rubric; instructor's scored feedback (private course file)
Historical notebook and original course record	notebook/milestone1_reproducibility.ipynb; artifacts/milestone1/2026-09-22/{progress_snapshot,evaluation_snapshot}.json; original v2 PDF
Clean dataset and frozen control	artifacts/milestone1/2026-09-17/approved_candidates.jsonl (SHA-256 f0d9324c68e7865f87cb1da3351d059b25ed08d2e08b9548073860ca7c9f702c); data/curated/student-authored-control-set-2026-09-22.jsonl (SHA-256 a1deb52304d7bce939593a228836debe2359488159e6e3b112e9d48ac7bea55c)
Human pilot and quality flags (historical archive only)	artifacts/milestone1/human_scoring/2026-09-24/human_generated_sample_scores.json; artifacts/milestone1/quality_review/2026-09-24/INAI6986_1999Record_QualityReview_HumanTemplate.xlsx; these are not current matched-run metrics.
Current eight-run Gemma/Muse result matrix	artifacts/model_comparison/2026-09-25/results.jsonl (28 epoch records) and run_log.csv (eight current runs); Section 4 and Appendix E summarize all current metrics.
Gemma and Muse per-run completion artifacts	artifacts/model_comparison/2026-09-25/training_results/ and muse-runs/; all final states verified complete, best checkpoints selected by validation loss, no test/control rows used.
Legacy Muse provenance (excluded)	artifacts/model_comparison/2026-09-25/historical-run-results.json and legacy-muse-comparability-note.json; retained only as history and not included in current results tables/plots.
Nemotron/Ornith architecture-family review	artifacts/model_comparison/2026-09-25/architecture_backward_gate_review_2026-09-28.json; official sources recorded there; exact local backward/memory/checkpoint gates remain open.

Selected Technical Sources

Alabama Medicaid Administrative Code:
https://medicaid.alabama.gov/content/9.0_Resources/9.2_Administrative_Code.aspx

Alabama Medicaid Provider Manuals:
https://medicaid.alabama.gov/content/Gated/7.6.1G_Provider_Manuals.aspx

LoRA method: Hu et al., https://arxiv.org/abs/2106.09685. Candidate model identities and exact pinned revisions are recorded in artifacts/model_comparison/2026-09-25/source-state-inventory.json; no third-party performance claim is inferred from a model card.

Appendix E — Complete Current Gemma and Muse Epoch Results

This table contains the 28 current training/validation records from results.jsonl and excludes the pre-amendment September 16–18 Muse runs. The eight runs use document-balanced nested train prompt IDs (N=200/509/750/1,600) and the same fixed 124 validation IDs for both model families. Results are teacher-forced causal next-token metrics only. A dash means that metric was not recorded for the checkpoint, not zero; for example, the partial-epoch Gemma N=509 step-200 checkpoint has validation but no train-split metric. The step-200 Gemma N=509 row is a partial-epoch work-window checkpoint before the run resumed; the final completed run ended at step 320.

Model / N	Epoch / Step	Train Loss	Train Next-Token Acc.	Val. Loss	Val. Next-Token Acc.	Scored Tokens: Train / Val.
Gemma 4; N=200	1.000 / 25	1.1405	74.60%	—	—	— / —
Gemma 4; N=200	2.000 / 50	0.7575	82.99%	0.7751	82.15%	28,056 / 18,125
Gemma 4; N=200	3.000 / 75	0.6893	84.40%	0.7044	83.45%	28,056 / 18,125
Gemma 4; N=509	1.000 / 64	0.6592	84.73%	0.6508	84.19%	73,166 / 18,125
Gemma 4; N=509	2.000 / 128	0.5430	86.60%	0.6060	84.93%	73,166 / 18,125
Gemma 4; N=509	3.000 / 192	0.4433	88.52%	0.6067	85.19%	73,166 / 18,125
Gemma 4; N=509	3.126 / 200	—	—	0.6178	85.25%	— / 18,125
Gemma 4; N=509	4.000 / 256	0.3688	90.06%	0.6321	85.24%	73,166 / 18,125
Gemma 4; N=509	5.000 / 320	0.3373	90.81%	0.6636	85.03%	73,166 / 18,125
Gemma 4; N=750	1.000 / 94	0.6182	85.26%	0.6199	84.68%	109,023 / 18,125
Gemma 4; N=750	2.000 / 188	0.4791	87.69%	0.5829	85.58%	109,023 / 18,125
Gemma 4; N=750	3.000 / 282	0.3361	90.63%	0.6205	85.37%	109,023 / 18,125
Gemma 4; N=750	4.000 / 376	0.2192	93.62%	0.6946	85.13%	109,023 / 18,125
Gemma 4; N=1,600	1.000 / 200	0.5400	86.69%	0.5751	85.79%	232,550 / 18,125
Gemma 4; N=1,600	2.000 / 400	0.3832	89.63%	0.5750	85.88%	232,550 / 18,125
Gemma 4; N=1,600	3.000 / 600	0.2435	93.05%	0.6180	85.65%	232,550 / 18,125
Muse Glimmer; N=200	1.000 / 25	0.5252	86.63%	0.5853	85.10%	24,507 / 15,903
Muse Glimmer; N=200	2.000 / 50	0.3505	90.32%	0.6073	84.98%	24,507 / 15,903
Muse Glimmer; N=200	3.000 / 75	0.2376	93.15%	0.6589	84.54%	24,507 / 15,903
Muse Glimmer; N=509	1.000 / 64	0.4638	87.86%	0.5727	85.44%	64,019 / 15,903
Muse Glimmer; N=509	2.000 / 128	0.2305	93.62%	0.6264	84.86%	64,019 / 15,903
Muse Glimmer; N=509	3.000 / 192	0.0856	97.64%	0.7456	84.70%	64,019 / 15,903
Muse Glimmer; N=750	1.000 / 94	0.4532	88.04%	0.5725	85.44%	95,431 / 15,903
Muse Glimmer; N=750	2.000 / 188	0.2137	93.82%	0.6279	85.32%	95,431 / 15,903
Muse Glimmer; N=750	3.000 / 282	0.0846	97.70%	0.7447	84.80%	95,431 / 15,903
Muse Glimmer; N=1,600	1.000 / 200	0.4067	89.15%	0.5787	85.54%	203,025 / 15,903
Muse Glimmer; N=1,600	2.000 / 400	0.1969	94.23%	0.6054	85.41%	203,025 / 15,903
Muse Glimmer; N=1,600	3.000 / 600	0.0877	97.56%	0.7249	84.72%	203,025 / 15,903